

1. Calcule la siguiente integral:

$$\int e^x \left[\frac{1-5x+x^2+\operatorname{sen}(x)-5x\cos(x)+x^2\cos(x)}{1+\cos(x)} \right] dx$$

$$\begin{aligned} \int e^x \left(\frac{-5x(1+\cos(x)) + x^2(1+\cos(x)) + (1+\operatorname{sen}(x))}{1+\cos(x)} \right) dx &= \int (x^2-5x)e^x dx + \int (1+\operatorname{sen}(x))e^x dx \\ &= (x^2-5x)e^x - (5x-5)e^x + 2e^x + e^x \tan x + C \\ * \int (1+\operatorname{sen}(x))e^x dx &= \int \frac{e^x}{1+\cos(x)} dx + \int \frac{e^x \operatorname{sen}(x) dx}{1+\cos(x)} = e^x \tan x \\ \text{II} = \int \frac{e^x}{2\cos^2 \frac{x}{2}} dx &= \frac{1}{2} \int e^x \sec^2 \frac{x}{2} dx = e^x \tan \frac{x}{2} - \int e^x \tan x dx \\ e^x = u \quad \frac{d}{dx} e^x = du \quad \frac{d}{dx} \tan \frac{x}{2} = \frac{1}{2} \sec^2 \frac{x}{2} \\ \text{I} = \int \frac{e^x \sec^2 \frac{x}{2}}{\sec^2 \frac{x}{2}} dx &= \int e^x \tan x dx \end{aligned}$$

2. Encontrar K de la siguiente formula de recurrencia:

$$I = \int x^m (a + bx^2) dx = \frac{x^{m+1} (a + bx^2)^2}{b(m+3)} - K \int x^{m-2} (a + bx^2) dx$$

$$\begin{aligned} \text{I} = \int x^m (a + bx^2) dx &= \int a x^m dx + b \int x^{m+2} dx = \frac{a x^{m+1}}{m+1} + \frac{b x^{m+3}}{m+3} \\ \text{I} = \frac{x^{m+1} (a + bx^2)^2}{b(m+3)} - K \int a x^{m-2} dx - K \int b x^{m+2} dx &= \frac{x^{m+1} (a + bx^2)^2}{b(m+3)} - \frac{a x^{m-1}}{m-1} - \frac{b x^{m+3}}{m+3} \\ &= \frac{x^{m+1} (a^2 + 2abx^2 + b^2 x^4)}{b(m+3)} - \frac{a x^{m-1}}{m-1} - \frac{b x^{m+3}}{m+3} \\ &= \left(\frac{a^2}{b(m+3)} - \frac{a}{m-1} \right) x^{m+1} + \left(\frac{2a}{m+3} - \frac{b}{m+1} \right) x^{m+3} + \frac{b x^{m+3}}{m+3} \\ \frac{a^2}{b(m+3)} - \frac{a}{m-1} &= 0 \quad \text{cumple} \\ \frac{2a}{m+3} - \frac{b}{m+1} &= K \end{aligned}$$

$$K = \frac{a(m-1)}{b(m+3)}$$

3. Calcule la siguiente integral:

$$\int \frac{\tan^2(x) - 3 \tan(x) + 7}{\sqrt{2 \operatorname{sen}^2(x) \cos^2(x) + 4 \operatorname{sen}(x) \cos^3(x) + 5 \cos^4(x)}} dx$$

$$\begin{aligned} \int \frac{+\tan^2 x - 3 \tan x + 7}{\sqrt{2 \operatorname{sen}^2 x + 4 \tan x + 5}} dx &= \int \frac{(\tan^2 x - 3 \tan x + 7) \sec^2 x}{\sqrt{2 \operatorname{sen}^2 x + 4 \tan x + 5}} dx = \int \frac{(u^2 - 3u + 7) du}{\sqrt{2u^2 + 4u + 5}} \\ \tan x = u \rightarrow \sec^2 x dx = du &= \frac{1}{2} \int \frac{\sqrt{2u^2 + 4u + 5} du - \frac{9}{4} \int \frac{(4u+4) du}{\sqrt{2u^2 + 4u + 5}} + 3 \int \frac{du}{\sqrt{u^2 + 1}^2} + \frac{5}{2} \\ * \int \frac{du}{\sqrt{u^2 + 1}^2} &= \operatorname{arcsen} \left(\frac{u}{1} \right) \\ \text{II} = \int \sqrt{2u^2 + 4u + 5} du &= \sqrt{2} \int \sqrt{u^2 + u + \frac{5}{2}} du = \sqrt{2} \int \frac{3}{4} \sec \theta \sec^3 \theta d\theta = \frac{\sqrt{2}}{4} \int \sec^4 \theta d\theta = \frac{\sqrt{2}}{4} \int \sec^2 \theta \sec^2 \theta d\theta \\ &= \frac{\sqrt{2}}{4} \int \sec^2 \theta d\theta \int \sec^2 \theta d\theta = \frac{\sqrt{2}}{4} \sec \theta \tan \theta - \frac{\sqrt{2}}{4} \int \sec \theta d\theta \\ &= \frac{\sqrt{2}}{4} \sec \theta \tan \theta - \frac{\sqrt{2}}{4} \int \sec \theta d\theta \\ \int \sec \theta d\theta &= \frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \end{aligned}$$

4. Calcule la siguiente integral:

$$\int \frac{\sec^2(x)}{(\tan(x) - 2) \sqrt{5 \tan(x) - \sec^2(x) - 3}} dx$$

$$\begin{aligned} \tan x = u \rightarrow \sec^2 x dx = du \\ \int \frac{du}{(u-2) \sqrt{5u - u^2 - 4}} &= \int \frac{du}{(u-2) \sqrt{\frac{5}{4} - (u-\frac{5}{2})^2}} = \int \frac{\frac{2}{5} \sec \theta d\theta}{\left(\frac{2}{5} \sec \theta + \frac{1}{5}\right) \frac{2}{5} \sec \theta} = \int \frac{2 d\theta}{3 \sec \theta + 1} = 2 \sqrt{2} \ln \left| \frac{\tan(\operatorname{arcsen}(\frac{2 \tan \theta - 1}{2})) - 2 \sqrt{2}}{\tan(\operatorname{arcsen}(\frac{2 \tan \theta - 1}{2})) + 2 \sqrt{2}} \right| + C \\ u = \frac{5}{2} \quad \frac{5}{2} - \frac{5}{2} = \frac{2}{5} \sec \theta \rightarrow \frac{2 \sqrt{2}}{5} = \sec \theta \\ du = \frac{2}{5} \sec^2 \theta d\theta \end{aligned}$$

$$* \int \frac{2 d\theta}{3 \sec \theta + 1} = 4 \int \frac{du}{6 \tan u + 1} = 4 \int \frac{\sec^2 u du}{1 + 6 \tan u + 3 \tan^2 u} = 4 \int \frac{d(\tan u)}{(3 \tan u + 1)^2} = 2 \sqrt{2} \ln \left| \frac{\tan(\frac{\theta}{3}) - 2 \sqrt{2}}{\tan(\frac{\theta}{3}) + 2 \sqrt{2}} \right|$$

5. Calcule la siguiente integral:

$$\int \frac{1}{\sqrt{x-1} \sqrt{2-x} [4\sqrt{x-1} + 3\sqrt{2-x}]} \cdot dx$$

$$\begin{aligned} \sqrt{x-1} \quad \frac{1}{\sqrt{2-x}} \quad \frac{dx}{2\sqrt{x-1}} = \sec \theta \quad \rightarrow \int \frac{2 \sec \theta d\theta}{\sec \theta (4 \sec \theta + 3 \sec \theta)} = \frac{2}{5} \int \frac{d\theta}{4 \sec \theta + 3} = \frac{2}{5} \int \frac{1}{\sec \theta (4 + 3 \cos \theta)} = \frac{2}{5} \ln \left| \frac{\sec \theta - 4}{\sec \theta + 4} \right| + C \\ = \frac{2}{5} \ln \left| \frac{\sec(\operatorname{arcsen}(\frac{1}{\sqrt{2-x}})) - 4}{\sec(\operatorname{arcsen}(\frac{1}{\sqrt{2-x}})) + 4} \right| + C \end{aligned}$$

6. a) Expresar como una integral definida el siguiente limite de

$$\text{una sumatoria } \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{n^2 + 8ni + 16i^2}{2n^3 + 12n^2i + 48ni^2 + 64i^3}$$

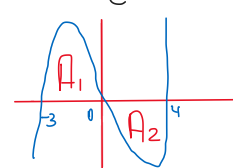
b) Mediante la definición de integral como el límite de

una sumatoria calcule el área de la región acotada por la

curva $y = x^3 - x^2 - 12x$ y el eje X

$$\begin{aligned} \text{a) } \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{(1+8(\frac{i}{n})+16(\frac{i}{n})^2)}{2+12(\frac{i}{n})+48(\frac{i}{n})^2+64(\frac{i}{n})^3} \cdot \frac{1}{n} &= \int_0^1 \frac{1+8x+16x^2}{2+12x+48x^2+64x^3} dx \\ \Delta x = \frac{1}{n} \quad x_i = \frac{i}{n} \quad \left. \begin{array}{l} \Delta x = \frac{1}{n} \\ x_i = 0 + i \Delta x \end{array} \right\} \begin{array}{l} a=0 \\ b=1 \end{array} \end{aligned}$$

$$\text{b) } A = x^3 - x^2 - 12x \rightarrow A_+ = A_1 + A_2 = \int_0^0 x^3 dx - \int_0^4 x^3 dx = \frac{0^4}{4} - \frac{4^4}{4} = -64$$



$$\text{I}_1 = \int_0^0 x^3 dx = 0$$

$$\text{I}_2 = \int_0^4 x^3 dx = \frac{4^4}{4} = 64$$

$$\text{I}_3 = \int_0^4 x^2 dx = \frac{4^3}{3} = \frac{64}{3}$$

$$\text{I}_4 = \int_0^4 x dx = \frac{4^2}{2} = 8$$

$$\text{I}_5 = \int_0^4 1 dx = 4$$

$$\text{I}_6 = \int_0^4 x^3 dx = \frac{4^4}{4} = 64$$

$$\text{I}_7 = \int_0^4 x^2 dx = \frac{4^3}{3} = \frac{64}{3}$$

$$\text{I}_8 = \int_0^4 x dx = \frac{4^2}{2} = 8$$

$$\text{I}_9 = \int_0^4 1 dx = 4$$

$$\text{I}_{10} = \int_0^4 x^3 dx = \frac{4^4}{4} = 64$$

$$\text{I}_{11} = \int_0^4 x^2 dx = \frac{4^3}{3} = \frac{64}{3}$$

$$\text{I}_{12} = \int_0^4 x dx = \frac{4^2}{2} = 8$$

$$\text{I}_{13} = \int_0^4 1 dx = 4$$

$$\text{I}_{14} = \int_0^4 x^3 dx = \frac{4^4}{4} = 64$$

$$\text{I}_{15} = \int_0^4 x^2 dx = \frac{4^3}{3} = \frac{64}{3}$$

$$\text{I}_{16} = \int_0^4 x dx = \frac{4^2}{2} = 8$$

$$\text{I}_{17} = \int_0^4 1 dx = 4$$

$$\text{I}_{18} = \int_0^4 x^3 dx = \frac{4^4}{4} = 64$$

$$\text{I}_{19} = \int_0^4 x^2 dx = \frac{4^3}{3} = \frac{64}{3}$$

$$\text{I}_{20} = \int_0^4 x dx = \frac{4^2}{2} = 8$$